

```

# 1. Draw a Scatter Plot using Matplotlib for X and y_reg. Add a
title, X-axis label, and Y-axis label.
# 2. Create a Linear Regression model and predict the output for X =
7.
# 3. Create a Logistic Regression model and predict the class for X =
4.
# 4. Create a KNN Classifier using k = 3 and predict the class for X =
5.
# 5. Create a Random Forest Classifier with 10 trees and predict the
class for X = 5.
# 6. Print the prediction of each model.

```

Note:- Handle the Null values before train the Model and also in Graphs etc.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression

```

```

X = [1, 2, 3, 4, 5, None, 7, 8, 9, 10, 11, 12, None, 14, 15, 16, 17,
18, 19, 20]
y_reg = [2, 4, 6, 8, 10, None, 14, 16, 18, 20, 22, 24, None, 28, 30,
32, 34, 36, 38, 40]
y_cls = [0, 0, 0, 0, 0, None, 0, 1, 1, 1, 1, 1, None, 1, 1, 1, 1, 1,
1, 1]

```

```

d1=pd.DataFrame(X,columns=['X'])
d2=pd.DataFrame(y_reg,columns=['y_reg'])

```

```

data=pd.concat([d1,d2],axis=1)
Data=data.dropna().copy()

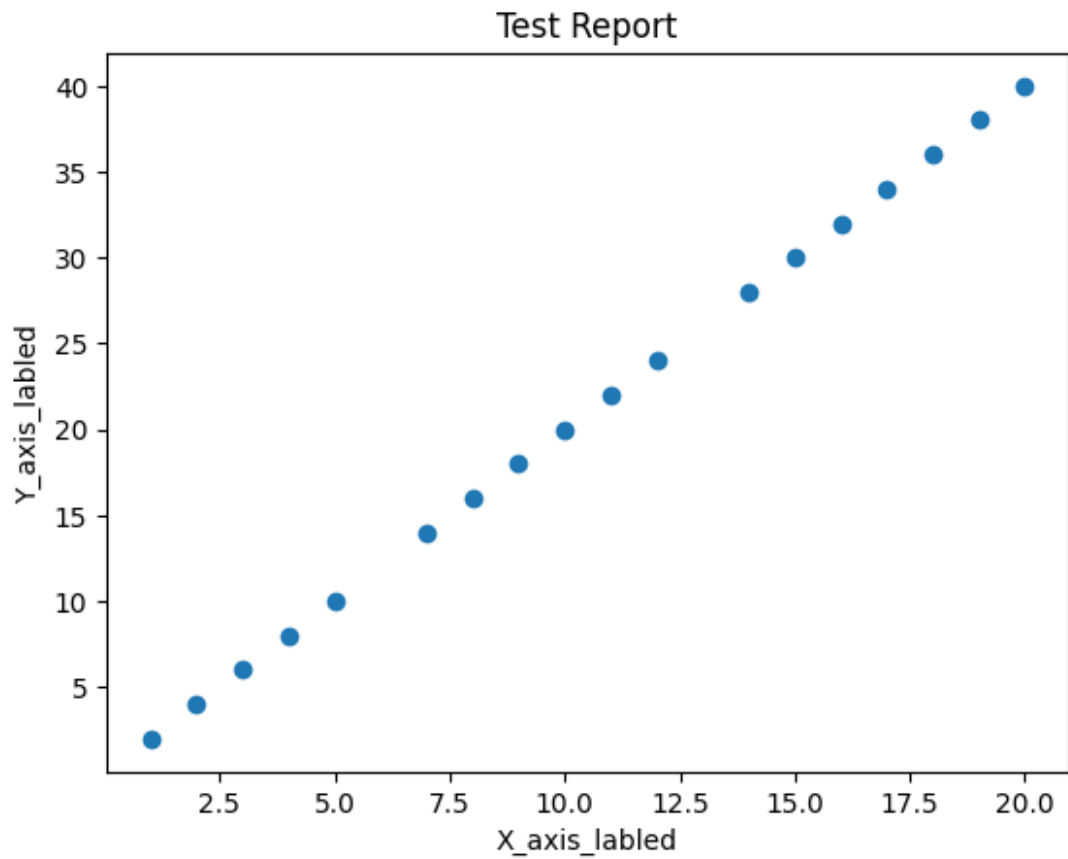
```

Data

```

plt.scatter(Data['X'], Data['y_reg'])
plt.title("Test Report")
plt.xlabel("X_axis_labled")
plt.ylabel("Y_axis_labled")
Text(0, 0.5, 'Y_axis_labled')

```



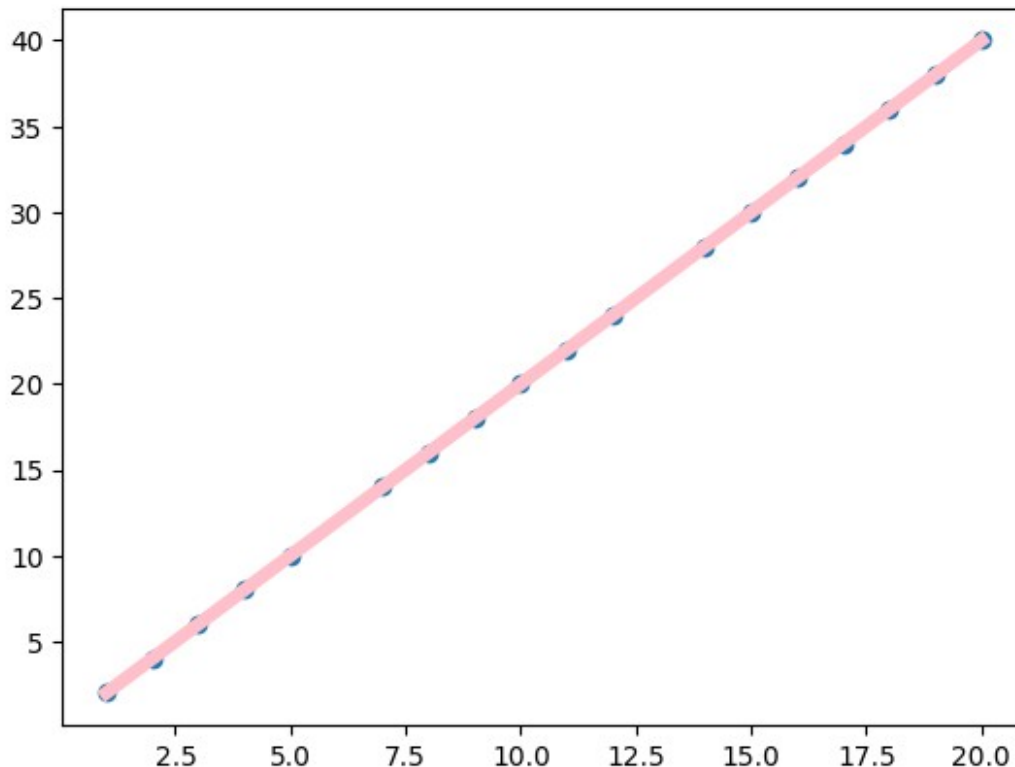
```
x=Data[['X']]
y=Data[['y_reg']]

regression=LinearRegression()
regression.fit(x,y)

LinearRegression()

y_pred=regression.predict(x)
plt.scatter(x,y)
plt.plot(x,y_pred,linewidth=5,color='pink')

[<matplotlib.lines.Line2D at 0x7a673f689880>]
```



```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y)
```

```
from sklearn.linear_model import LogisticRegression
```

```
logistic_model=LogisticRegression()
```

```
logistic_model.fit(x,y)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/
validation.py:1408: DataConversionWarning: A column-vector y was
passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
```

```
    y = column_or_1d(y, warn=True)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic
.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
    n_iter_i = _check_optimize_result(
```

```
LogisticRegression()

y_pred=logistic_model.predict(x_test)
y_pred
x_pred=logistic_model.predict(x_train)

from sklearn.metrics import accuracy_score,classification_report
print("Training Accuracy =",accuracy_score(y_train,x_pred))
print("Training Accuracy = ",accuracy_score(y_test,y_pred))

Training Accuracy = 0.46153846153846156
Training Accuracy = 1.0
```